Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Enterprise Java

Practical IX

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Date of Assignment: 12-9-2026	Date/Time of Submission: 12-9-2026

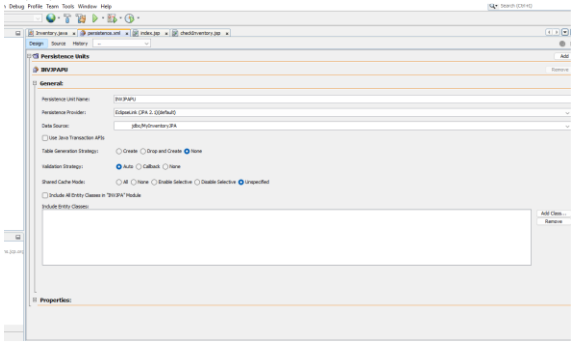
Aim:- Implement the following JPA applications.

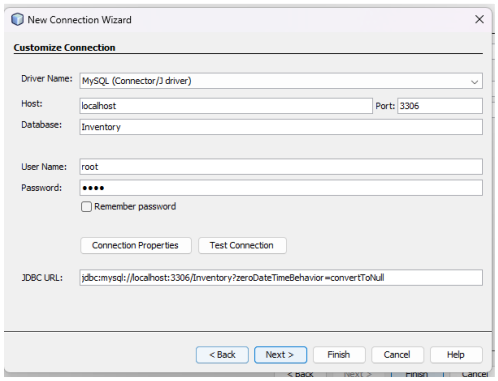
A. Develop a simple Inventory Application Using JPA.

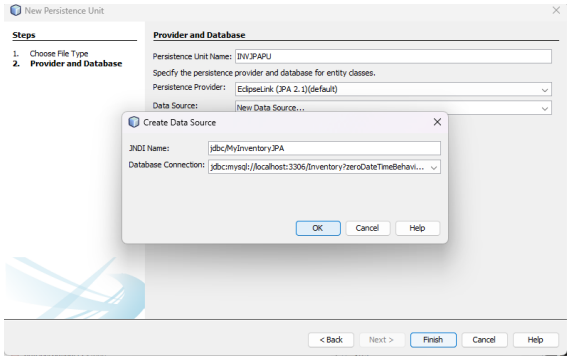
Code:-

Inventory.java

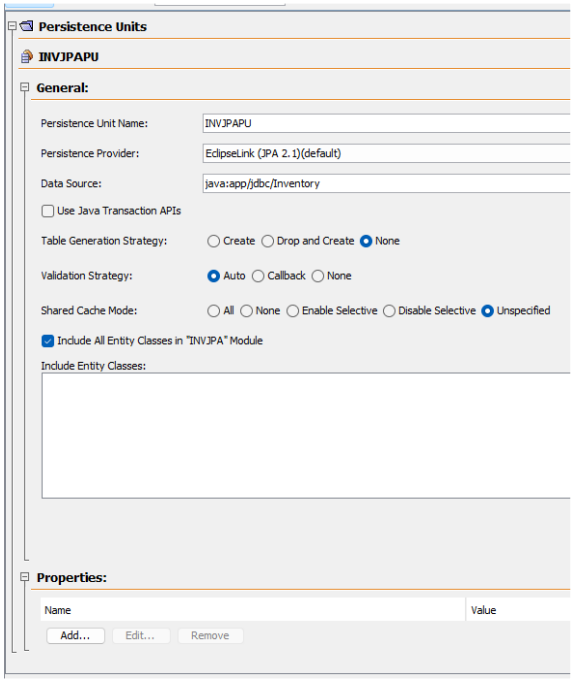
```
package myApp;
import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;
@Entity
@Table(name = "inventory")
public class Inventory {
   @Id
   @GeneratedValue(strategy =
GenerationType.IDENTITY)
   @Column(name = "itemid", unique = true, updatable =
false)
   private Integer itemid;
   @Column(name = "itemName")
   private String itemName;
   @Column(name = "description")
   private String description;
   @Column(name = "quantity")
   private Integer quantity;
   @Column(name = "addedDate")
   private String addedDate;
   public Inventory() {
   }
   public Integer getItemID() {
      return itemid;
   }
   public void setItemID(Integer itemid) {
      this.itemid = itemid;
   }
   public String getItemName() {
      return itemName;
   }
   public void setItemName(String itemName) {
      this.itemName = itemName;
   }
   public String getDescription() {
      return description;
   }
   public void setDescription(String description) {
      this.description = description;
   }
   public Integer getQuantity() {
      return quantity;
   }
   public void setQuantity(Integer quantity) {
      this.quantity = quantity;
   }
   public String getAddedDate() {
      return addedDate;
   }
   public void setAddedDate(String addedDate) {
      this.addedDate = addedDate;
   } }
```







Persistence .xml



Sql :

```
mysql> CREATE DATABASE Inventory;
Query OK, 1 row affected (0.01 sec)

mysql> USE Inventory;
Database changed
mysql>
mysql> CREATE TABLE inventory (
    ->      itemid INT PRIMARY KEY AUTO_INCREMENT,
    ->      itemName VARCHAR(50),
    ->      description VARCHAR(200),
    ->      quantity INT,
    ->      addedDate VARCHAR(40)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> USE Inventory;
Database changed
mysql> SHOW TABLES;
+---------------------+
| Tables_in_inventory |
+---------------------+
| inventory           |
+---------------------+
1 row in set (0.03 sec)
```

Index.jsp:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>

<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>Inventory</title>
</head>
<body>
<form action="checkInventory.jsp" method="post">
<table>
<tr>
    <td colspan="2" align="center">
        <b>Add the item in Inventory</b>
    </td>
</tr>
<tr>
    <td>Item Name:</td>
    <td>
        <input name="itemname" maxlength="25" size="50"/>
    </td>
</tr>
<tr>
    <td>Description:</td>
    <td>
        <textarea rows="5" cols="36" name="description"></textarea>
    </td>
</tr>
<tr>
    <td>Quantity:</td>
    <td>
        <input name="quantity" maxlength="4" size="5"/>
    </td>
</tr>
<tr>
    <td colspan="2">
        <input type="submit"
            name="AddToInventory"
            value="AddToInventory"/>
    </td>
</tr>
</table>
</form>
</body>
</html>
```

checkInventory.jsp:

```
<%@page import="java.util.Iterator"%>
<%@page import="javax.persistence.Persistence"%>
<%@page import="java.util.List"%>
<%@page import="myApp.Inventory"%>
<%@page import="javax.persistence.EntityTransaction"%>
<%@page import="javax.persistence.EntityManager"%>
```

```
<%@page
import="javax.persistence.EntityManagerFactory"%>
<%@page contentType="text/html" pageEncoding="UTF-
8"%>
<%!
   private EntityManagerFactory eMF;
   private EntityManager eM;
   private EntityTransaction eT;
   List<Inventory> inv;
%>
<%
   eMF =
Persistence.createEntityManagerFactory("INVJPAPU");
   eM = eMF.createEntityManager();
   String submit =
request.getParameter("AddToInventory");
   if (submit != null && submit.equals("AddToInventory"))
{
      try {
         String itemname =
request.getParameter("itemname");
         String description =
request.getParameter("description");
         String quantity1 = request.getParameter("quantity");
         int quantity = Integer.parseInt(quantity1);
         String addedDate = new java.util.Date().toString();
         Inventory inventory = new Inventory();
         inventory.setItemName(itemname);
         inventory.setDescription(description);
         inventory.setQuantity(quantity);
         inventory.setAddedDate(addedDate);
         eT = eM.getTransaction();
         eT.begin();
         eM.persist(inventory);
         eT.commit();
      } catch (RuntimeException e) {
         if (eT != null && eT.isActive()) {
            eT.rollback();
         }
         throw e;
      }
      response.sendRedirect("checkInventory.jsp");
      return;
   }
   inv = eM.createQuery(
      "SELECT i FROM Inventory i",
      Inventory.class
   ).getResultList();
   eM.close();
   eMF.close();
%>

<!DOCTYPE html>
<html>
<head>
   <meta http-equiv="Content-Type"
      content="text/html; charset=UTF-8">
   <title>Inventory</title>
</head>
<body>
<table border="1">
<tr>
   <td colspan="5" align="center">
      <b>
         Click <a href="index.jsp">here</a>
         to add items in the Inventory.
      </b>
   </td>
</tr>
<tr>
   <td>Item No</td>
   <td>Item Name</td>
   <td>Description</td>
   <td>Quantity</td>
   <td>Added Date & Time</td>
</tr>
<%
   Iterator<Inventory> iterator = inv.iterator();
   while (iterator.hasNext()) {
      Inventory objInv = iterator.next();
%>
<tr>
   <td><b><%=objInv.getItemID()%></b></td>
   <td><b><%=objInv.getItemName()%></b></td>
   <td><b><%=objInv.getDescription()%></b></td>
   <td><b><%=objInv.getQuantity()%></b></td>
   <td><b><%=objInv.getAddedDate()%></b></td>
</tr>
<%
   }
%>
</table>
</body>
</html>
```

Output:-

